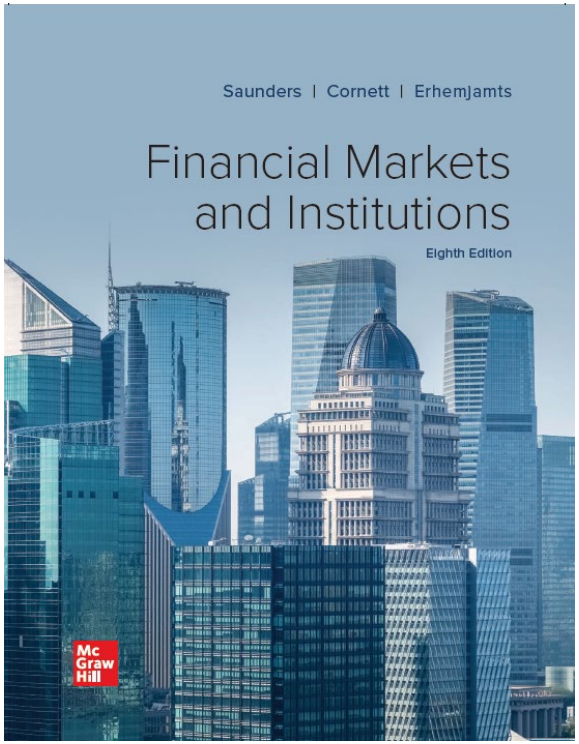
Chapter Two

Determinants of Interest Rates



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Interest Rate Fundamentals

- **Nominal interest rates** are the interest rates actually observed in financial markets
  - Directly affect the value (price) of most securities traded in the money and capital markets
  - Changes in interest rates influence the performance and decision making for individual investors, businesses, and governmental units

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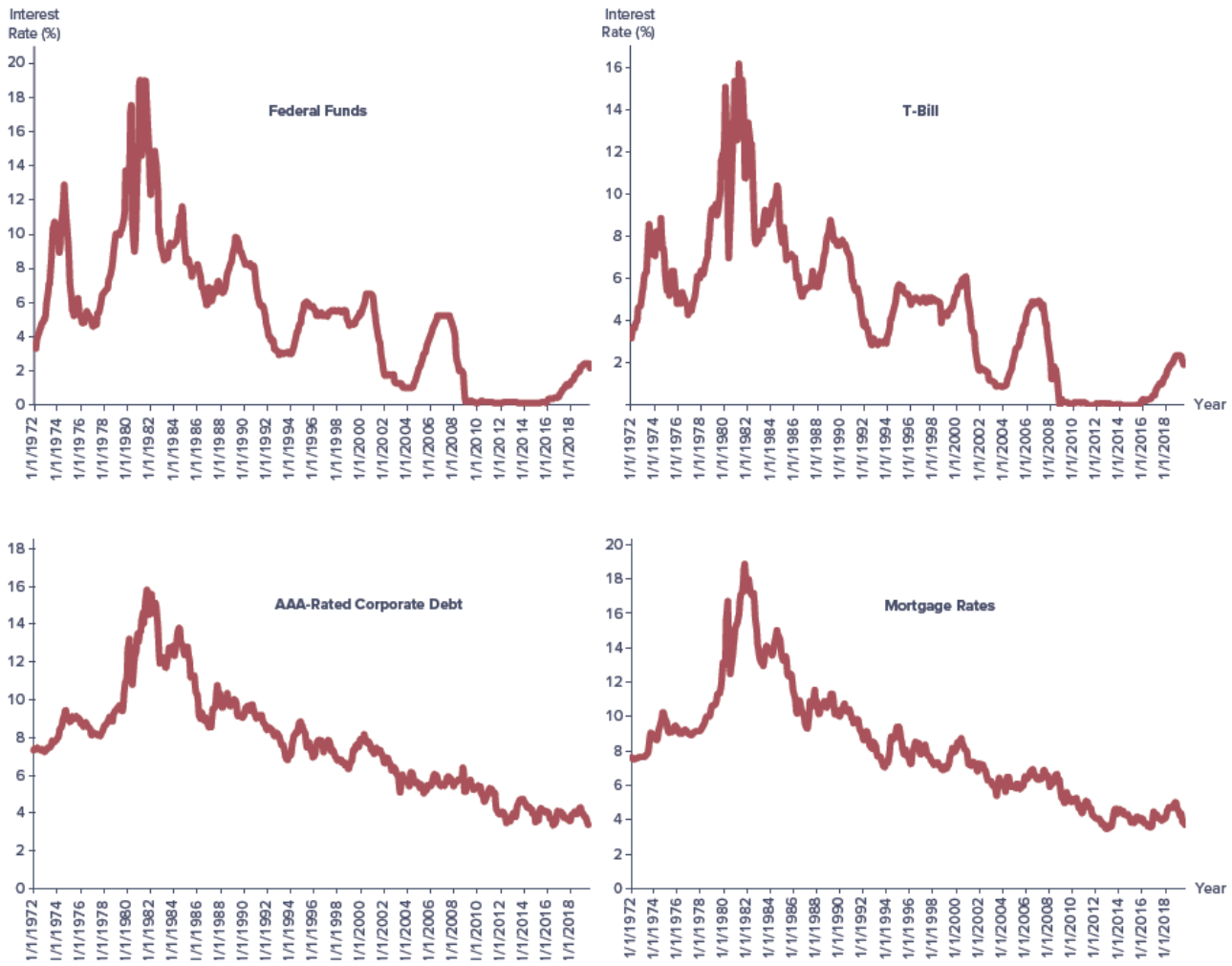
Key U.S. Interest Rates, 1972-2019



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Loanable Funds Theory

- Changes in interest rates impact security values
  - FIs spend much time and effort trying to identify factors that determine the level of interest rates at any moment in time, as well as what causes interest rate movements over time
- **Loanable funds theory** views equilibrium interest rates in financial markets as a result of the supply of and demand for loanable funds
  - Categorizes financial market participants – consumers, businesses, governments, and foreign participants – as net suppliers or demanders of funds

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Supply of Loanable Funds

- "Supply of loanable funds" describes funds provided to the financial markets by net suppliers of funds
  - Generally, the quantity of loanable funds supplied increases as interest rates rise
    - Household sector (consumer sector) is one of the largest suppliers of loanable funds in the U.S. ($84.66t in 2019)
    - Business sector often has excess cash that it can invest for short periods of time ($28.06t for nonfinancial and $98.47t for financial business in 2019)
    - Governments may supply loanable funds ($5.66t in 2019)
    - Foreign investors view U.S. markets as alternatives to their domestic financial markets ($27.20t in 2019)

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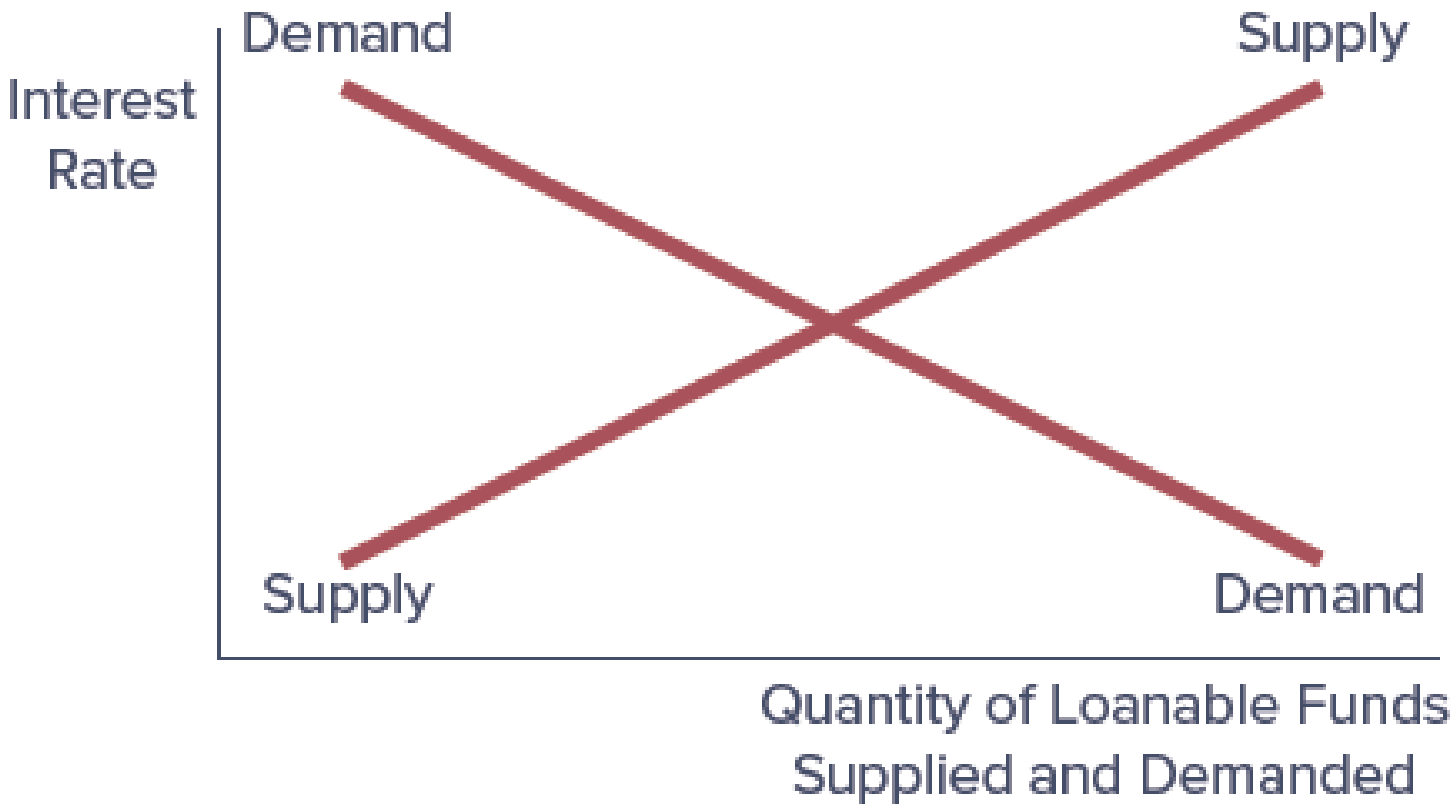
Supply of and Demand for Loanable Funds



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Demand for Loanable Funds

- "Demand for loanable funds" describes the total net demand for funds by fund users
  - In general, the quantity of loanable funds demanded is higher as interest rates fall
    - Household demand reflects financing purchases of homes, durable goods, and nondurable goods ($16.05t in 2019)
    - Businesses demand funds to finance investments in long-term assets and for short-term working capital needs ($66.46t for nonfinancial and $111.87t for financial in 2019)
    - Governments also borrow heavily ($28.86t in 2019)
    - Foreign participants, mostly from the business sector, borrow in U.S. financial markets ($20.81t in 2019)

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Factors That Cause the Supply and Demand Curves for Loanable Funds to Shift

- Factors that cause the supply curve of loanable funds to shift, at any given interest rate:
  1. As **wealth of fund suppliers** increases (decreases), the supply of loanable funds increases (decreases)
  2. As **risk of the financial security** increases (decreases), the supply of loanable funds decreases (increases)
  3. As **near-term spending needs** increase (decrease), the supply of loanable funds increases (decreases)
  4. When **monetary policy objectives** allow the economy to expand (restrict expansion), the supply of loanable funds increases (decreases)
  5. As **economic conditions** improve in a domestic (foreign) country, the supply of funds increases (decreases)

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Factors That Affect the Supply of and Demand for Loanable Funds for a Financial Security

Panel A: The supply of funds		
Factor	**Impact on Supply of Funds**	**Impact on Equilibrium Interest Rate***
Interest rate	Movement along the supply curve	Direct
Total wealth	Shift supply curve	Inverse
Risk of financial security	Shift supply curve	Direct
Near-term spending needs	Shift supply curve	Direct
Monetary expansion	Shift supply curve	Inverse
Economic conditions	Shift supply curve	Inverse

Panel B: The demand for funds		
Factor	**Impact on Demand for Funds**	**Impact on Equilibrium Interest Rate**
Interest rate	Movement along the demand curve	Direct
Utility derived from asset purchased with borrowed funds	Shift demand curve	Direct
Restrictiveness of nonprice conditions	Shift demand curve	Inverse
Economic conditions	Shift demand curve	Direct

*A "direct" impact on equilibrium interest rates means that as the "factor" increases (decreases) the equilibrium interest rate increases (decreases). An "inverse" impact means that as the factor increases (decreases) the equilibrium interest rate decreases (increases).

Factors That Cause the Supply and Demand Curves for Loanable Funds to Shift (Continued)

- Factors that cause the demand curve for loanable funds to shift include the following:
  1. As the **utility derived from an asset purchased with borrowed funds** increases (decreases), the demand for loanable funds increases (decreases)
  2. As the **restrictiveness of nonprice conditions on borrowed funds** decreases (increases), the demand for loanable funds increases (decreases)
     - Nonprice conditions may include fees, collateral, or requirements or restrictions on the use of funds (i.e., restrictive covenants)
  3. When domestic **economic conditions** result in a period of growth (stagnation), the demand for funds increases (decreases)

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Determinants of Interest Rates for Individual Securities: Inflation

- **Inflation** is the continual increase in the price level of a basket of goods and services
  - The higher the level of actual or expected inflation, the higher will be the level of interest rates
  - In the U.S., inflation is measured using indexes
    - Consumer price index (*CPI*)
    - Producer price index (*PPI*)
- Annual inflation rate using the CPI index between years *t* and *t*+1 would be equal to:

$$\text{Inflation } (IP) = \frac{CPI_{t+1} - CPI_t}{CPI_t} \times 100$$

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Determinants of Interest Rates for Individual Securities: Real Risk-Free Rate

- A **real risk-free rate** is the interest rate that would exist on a risk-free security if no inflation were expected over the holding period of a security
  - The higher society's preference to consume today, the higher the real risk-free rate (*RFR*)

- Relationship among the real risk-free rate (*RFR*), the expected rate of inflation [*E(IP)*], and the nominal interest rate (*i*) is referred to as the Fisher effect
  - The Fisher effect is often written as the following:

$$i = RFR + E(IP)$$

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Determinants of Interest Rates for Individual Securities: Default Risk

- **Default risk** is the risk that a security issuer will fail to make its promised interest and principal payments to the buyer of a security
  - The higher the default risk, the higher the interest rate that will be demanded by the buyer of the security to compensate him or her for this default (or credit) risk exposure
- Difference between a quoted interest rate on a security (security *j*) and a Treasury security with similar maturity, liquidity, tax, and other features (such as callability or convertibility) is called a *default* or *credit risk premium* (*DRPj*)

$$DRP_j = i_{jt} - i_{Tt}$$

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Determinants of Interest Rates for Individual Securities: Liquidity Risk

- **Liquidity risk** is the risk that a security can be sold at a predictable price with low transaction costs on short notice
  - A highly liquid asset is one that can be sold at a predictable price with low transaction costs, and thus can be converted into its full market value at short notice
  - If a security is illiquid, investors add a liquidity risk premium (LRP) to the interest rate on the security that reflects its relative liquidity
    - LRP might also be thought of as an “illiquidity” premium
  - LRP may also exist if investors dislike long-term securities because their prices (present values) are more sensitive to interest rate changes than short-term securities

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Determinants of Interest Rates for Individual Securities: Special Provisions or Covenants

- Special provisions or covenants that may be written into the contract underlying a security also affect the interest rates on different securities
  - Some of these provisions include the security's taxability, convertibility, and callability
    - For investors, interest payments on municipal securities are free of federal, state, and local taxes
    - A convertible (special) feature of a security offers the holder the opportunity to exchange one security for another type of the issuer's securities at a preset price
  - In general, special provisions that provide benefits to the security holder (e.g., tax-free status and convertibility) are associated with lower interest rates

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Determinants of Interest Rates for Individual Securities: Term to Maturity

- The **term structure of interest rates** is a comparison of market yields on securities, assuming all characteristics *except* maturity are the same
  - Change in required interest rates as the maturity of a security changes is called the maturity premium (MP)
    - The MP can be positive, negative, or zero

- The following general equation can be used to determine the factors that functionally impact the fair interest rate (i_j^*) on an individual (jth) financial security:

$$i_j^* = f(IP, RFR, DRP_j, LRP_j, SCP_j, MP_j)$$

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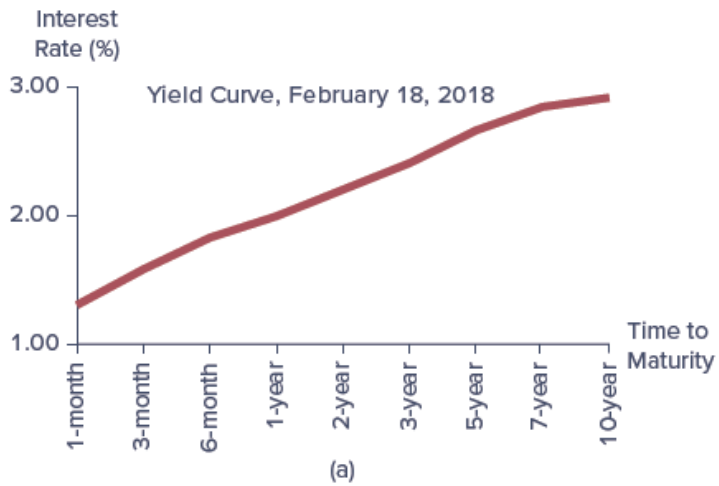
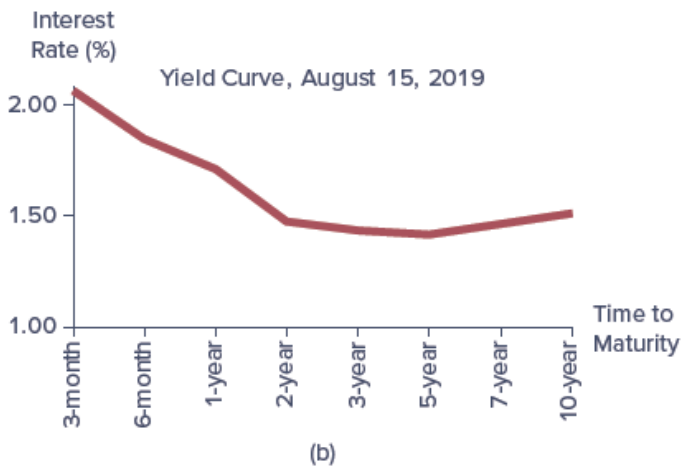
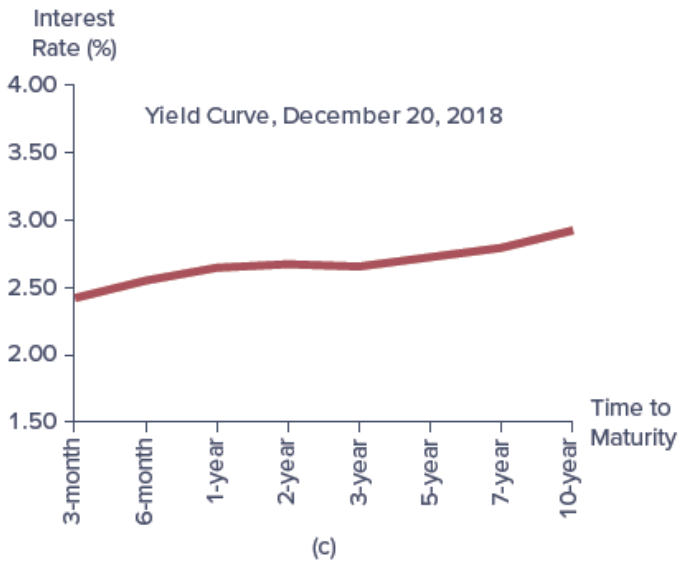
Common Shapes for Yield Curves on Treasury Securities







Term Structure of Interest Rates

- Relationship between a security's interest rate and its remaining term to maturity (i.e., the term structure of interest rates) can take a number of different shapes

- Explanations for the shape of the yield curve fall predominately into three theories:
  1. Unbiased expectations theory
  2. Liquidity premium theory
  3. Market segmentation theory

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Explanations for the Shape of the Term Structure of Interest Rates

Unbiased expectations theory—at any given point in time, the yield curve reflects the market's current expectations of future short-term rates. According to the unbiased expectations theory, the return for holding a four-year bond to maturity should equal the expected return for investing in four successive one-year bonds (as long as the market is in equilibrium).

Liquidity premium theory—long-term rates are equal to geometric averages of current and expected short-term rates, plus liquidity risk premiums that increase with the security's maturity. Longer maturities on securities mean greater market and liquidity risk. So, investors will hold long-term maturities only when they are offered at a premium to compensate for future uncertainty in the security's value. The liquidity premium increases as maturity increases.

Market segmentation theory—assumes that investors do not consider securities with different maturities as perfect substitutes. Rather, individual investors and FIs have preferred investment horizons (habitats) dictated by the nature of the liabilities they hold. Thus, interest rates are determined by distinct supply and demand conditions within a particular maturity segment (e.g., the short end and long end of the bond market).

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Unbiased Expectations Theory

- At a given point in time, the yield curve reflects the market's current expectations of future short-term rates

$$(1 + {}_1R_N)^N = (1 + {}_1R_1)(1 + E({}_2r_1)) \ldots (1 + E({}_Nr_1))$$

therefore:

$${}_1R_N = \left[(1 + {}_1R_1)(1 + E({}_2r_1)) \ldots (1 + E({}_Nr_1))\right]^{1/N} - 1$$

where

$${}_1R_N = \text{Actual } N\text{-period rate today (i.e., the first day of year 1)}$$
$$N = \text{Term to maturity, } N = 1, 2, \ldots, 4, \ldots$$
$${}_1R_1 = \text{Actual current one-year rate today}$$
$$E({}_ir_1) = \text{Expected one-year rates for years, } i = 2, 3, 4, \ldots, N \text{ in the future}$$

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Liquidity Premium Theory

- A weakness of the unbiased expectations theory is that it assumes that investors are risk neutral
- Liquidity premium theory is an extension of the unbiased expectations theory
  - Based on the idea that investors will hold long-term maturities only if they are offered at a premium to compensate for future uncertainty in a security's value, which increases with an asset's maturity

$$ {}_1R_N = [(1 + {}_1R_1)(1 + E({}_2r_1) + L_2) \ldots (1 + E({}_Nr_1) + L_N)]^{1/N} - 1 $$

where

L_t = Liquidity premium for a period t

$L_2 < L_3 < \ldots L_N$

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Market Segmentation Theory

- Market segmentation theory argues that individual investors and FIs have specific maturity preferences, and to get them to hold securities with maturities other than their most preferred requires a higher interest rate (maturity premium)
  - Does not consider securities with different maturities as perfect substitutes
  - Individual investors and FIs have preferred investment horizons (habitats) dictated by the nature of the liabilities they hold (i.e., investors have complete risk aversion for securities outside their maturity preferences)

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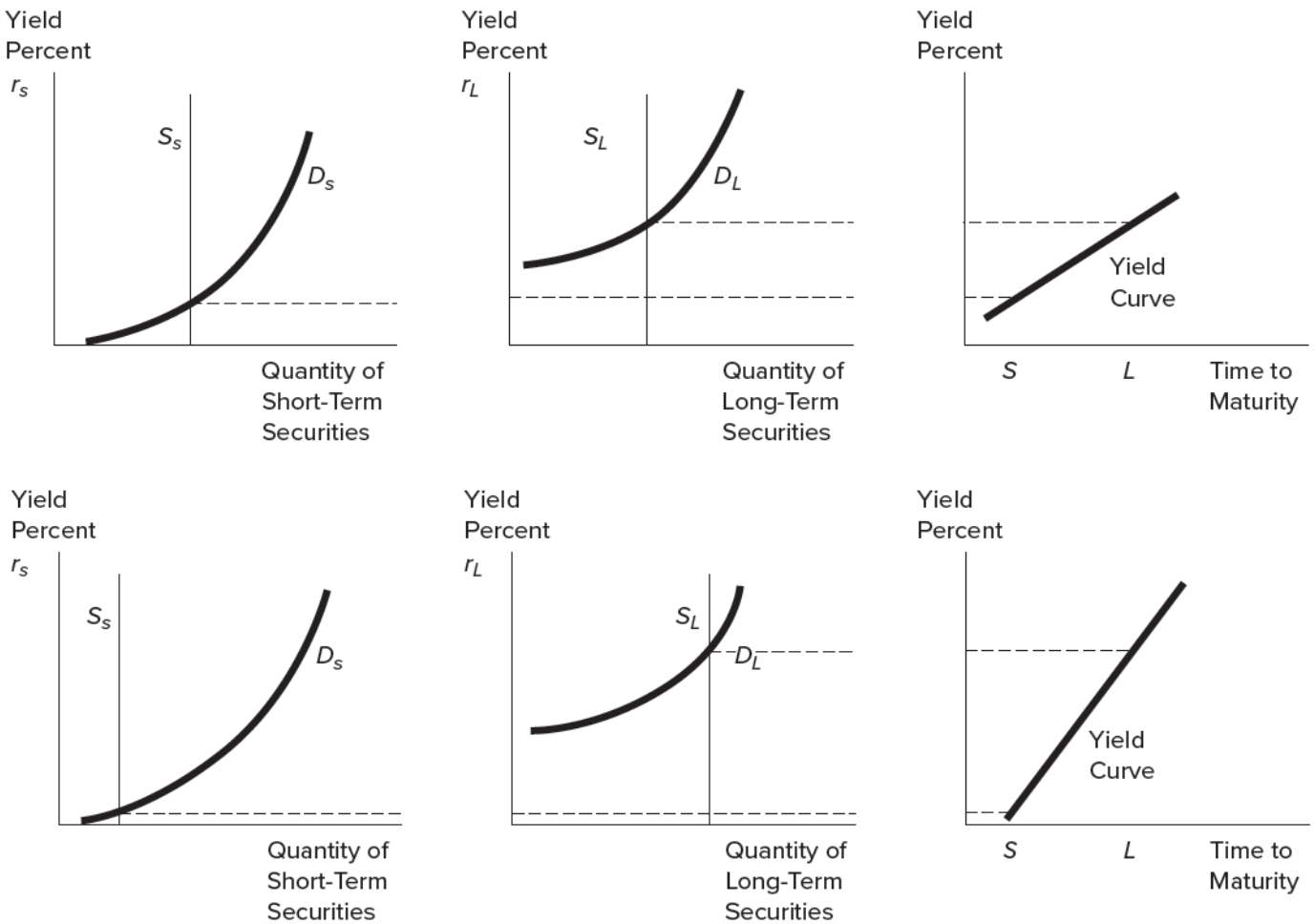
Market Segmentation and Determination of the Slope of the Yield Curve



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Time Value of Money

- Time value of money is the basic notion that a dollar received today is worth more than a dollar received at some future date
- Two forms of time value of money calculations are commonly used in finance for security valuation purposes:
  1. Value of a lump sum
     - A **lump sum payment** is a single cash payment received at the beginning or end of some investment horizon
  2. Value of annuity payments
     - **Annuity** payments are a series of equal cash flows received at fixed intervals over the entire investment horizon

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Lump Sum Valuation

- Present Value of a Lump Sum

$$PV = FV_t/(1+r)^t$$

- Future Value of a Lump Sump

$$FV_t = PV(1+r)^t$$

where

PV = Present value of cash flows
FV_t = Future value of cash flows (lump sum) received in t periods
r = Interest rate earned per period on an investment (equals the nominal annual interest rate, i, divided by the number of compounding periods per year—for example, daily, weekly, monthly, quarterly, semiannually)
t = Number of compounding periods in the investment horizon (equals the number of years in the investment horizon times the number of compounding periods per year)

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Annuity Valuation

- Present Value of an Annuity

$$PV = PMT \times \left[\frac{1 - \frac{1}{(1+r)^t}}{r} \right]$$

where

PMT = Periodic annuity payment received during an investment horizon

- Future Value of an Annuity

$$FV_t = PMT \times \left[\frac{(1+r)^t - 1}{r} \right]$$

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